BATCH NO:MP-AIDS-17

REAL TIME SUSPICIOUS ACTIVITY DETECTION USING CNN AND LSTM

MAJOR PROJECT SUPPORTING DOCUMENTS AND EVALUATION RECORDS

of

ANITHA D	(22UEAD0002)	**(VTU21542)**
MUTHARASI A	(22UEAD0038)	**(VTU22774)**

Under the guidance of
Mrs.K.Priya,M.E.,
ASSISTANT PROFESSOR

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
SCHOOL OF COMPUTING

VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF SCIENCE AND TECHNOLOGY

(Deemed to be University Estd u/s 3 of UGC Act, 1956)
Accredited by NAAC with A++ Grade
CHENNAI 600 062, TAMILNADU, INDIA

April 2026

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April 2026

VERIFICATION SHEET

This is to certify that the **PROJECT** titled **"REAL TIME SUSPICIOUS ACTIVITY DETECTION USING CNN AND LSTM"** by ANITHA D (22UEAD0002), MUTHARASI A (22UEAD0038) has been verified, and all the documents have been reviewed and approved as per the academic requirements.

Signature of Supervisor
Mrs.K.Priya
Assistant Professor
Artificial Intelligence and Data Science
School of Computing
Vel Tech Rangarajan Dr. Sagunthala R&D
Institute of Science and Technology

Signature of the Major Project Coordinator
Dr. M. Sam Navin
Assistant Professor
Artificial Intelligence and Data Science
School of Computing
Vel Tech Rangarajan Dr. Sagunthala R&D
Institute of Science and Technology

Signature of Head
Dr. P. Santhi
Professor & Head
Artificial Intelligence and Data Science
School of Computing
Vel Tech Rangarajan Dr. Sagunthala R&D
Institute of Science and Technology

Signature of the Dean
Dr. S P. Chokkalingam
Professor & Dean
School of Computing
Vel Tech Rangarajan Dr. Sagunthala R&D
Institute of Science and Technology

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1. SUPERVISOR SELECTION FORM

Vel Tech
Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)

NAAC A++

Recognized as CATEGORY-1 DEEMED TO BE UNIVERSITY by the UGC University Grants Commission

Batch No:MP-AIDS-~~18~~ 17

School of Computing

Department of Artificial Intelligence and Data Science

10214AD701- MAJOR PROJECT

Academic Year: 2025-2026

Major Project Supervisor Selection Form

I have read and understood the guidelines outlined under the B.Tech. VTR UGE-21 Regulations. Below are the details of our Major Project:

PROJECT DETAILS:

TITLE OF THE PROJECT: REAL -TIME SUSPICIOUS ACTIVITY DETECTION USING CNN - LSTM ~~AND ANOMALY DETECTION~~

DOMAIN OF THE PROJECT: ARTIFICIAL INTELLIGENCE

SUSTAINABLE DEVELOPMENT GOAL (SDG) CATEGORY: SDG 11 - SUSTAINABLE CITIES AND COMMUNITIES

TYPE OF THE PROJECT: AI - BASED SURVEILLANCE SYSTEM

I hereby declare that I will strictly adhere to all procedures and guidelines associated with the Major Project. I further affirm that the project will be executed independently by me, under the supervision of the assigned internal guide.

STUDENT DETAILS:

S.No	VTU No	Register No	Name of the Student	Signature
1.	22774	22UEAD0038	MUTHARASI·A	A·M
2.	21542	22UEAD0002	ANITHA·D	D·A
3.				

SUPERVISOR DETAILS:

TTS No	Name of the Supervisor	Designation	Research Area	Signature
3328	Mrs·K·PRIYA	ASSISTANT PROFESSOR	Machine Learning	

OFFICE DETAILS:

(Signature of the Project Coordinator-AIDS)

(Signature of the HoD-AIDS)

2. ABSTRACT FORM

Vel Tech
Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)

NAAC A++

Recognised as CATEGORY-1 DEEMED TO BE UNIVERSITY by the UGC University Grants Commission

Batch No: MP-AIDS- 17

School of Computing

Department of Artificial Intelligence and Data Science

10214AD701- MAJOR PROJECT

Academic Year: 2025-2026

Major Project - Abstract Form

STUDENT DETAILS:

S.No	VTU No	Register No	Name of the Student
1.	VTU22774	22UEAD0038	MUTHARASI A
2.	VTU21542	22UEAD0002	ANITHA D
3.			

TITLE & ABSTRACT:

Title: Real-Time Suspicious Activity Detection Using CNN-LSTM ~~and Anomaly Detection~~

Abstract: SafeCrowd Vision is an AI-based real-time surveillance system designed to enhance safety in densely crowded public spaces such as railway stations, markets, religious gatherings, and stadiums. Traditional monitoring methods often depend on human vigilance, which can be unreliable in high-density environments due to fatigue and oversight. To overcome these limitations, this project leverages advanced deep learning techniques to automatically detect suspicious activities and potential theft, significantly improving situational awareness and response capabilities.

The system combines Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks to analyze both spatial and temporal aspects of crowd behavior. CNNs extract key visual features from video frames—such as people, objects, and movements—while LSTMs analyze sequences of frames to detect anomalies like sudden running, tailgating, object grabbing, or unauthorized access. To further enhance accuracy, the system uses anomaly detection to compare real-time behaviors with learned normal patterns and integrates object tracking algorithms like YOLO and DeepSORT for identity tracking across frames.

SafeCrowd Vision operates in real time and issues automatic alerts to security personnel via mobile devices or dashboards when threats are detected, ensuring quick and proactive responses. Its flexibility allows integration with fixed CCTVs, drones, or mobile surveillance units. Trained on diverse datasets containing real and synthetic suspicious activities, the system is robust and adaptable to various environments, offering a scalable and intelligent solution to modern crowd safety challenges.

Keywords: Suspicious Activity Detection, Crowd Surveillance, Deep Learning, Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), Anomaly Detection, Computer Vision, Theft Detection, Real-Time Monitoring, Smart Surveillance, Behavior Analysis.

APPROVED BY:

Supervisor	Project Coordinator - AIDS	HOD-AIDS
(Mrs.K.Priya,M.E& TTS 3328)	*(Dr. M. Sam Navin)*	*(Dr. P. Santhi)*

3. ZEROTH REVIEW VALUATION

Vel Tech Rangarajan Dr. Sagunthala

Batch No: MP-AIDS-~~18~~ 17

School of Computing

Department of Artificial Intelligence and Data Science

10214AD701- MAJOR PROJECT

Academic Year: 2025-2026

ZEROTH REVIEW – VALUATION SHEET

PROJECT DETAILS:

TITLE OF THE PROJECT: Real - Time Suspicious Activity detection using CNN - LSTM

DOMAIN OF THE PROJECT: Artificial Intelligence

SUSTAINABLE DEVELOPMENT GOAL (SDG) CATEGORY: SDG11 - Sustainable cities and communities.

TYPE OF THE PROJECT: AI - Based Surveillance System

DATE OF THE ZEROTH REVIEW: 23.08.2025

STUDENT DETAILS:

S.No	VTU No	Register No	Name of the Student	Signature
1.	21542	22VEAD0002	Anitha. D	Anitha.
2.	22774	22VEAD0038	Mutharasi - A	Mut
3.				

SUPERVISOR REVIEW:

TTS No	Name of the Supervisor	Designation	Signature
3328	Mrs. K. Priya	Assistant prof	

PRESENTATION DELIVERED TO SUPERVISOR: YES/ NO

REVIEW COMMENTS FROM SUPERVISOR:

Explained well about the proposed system.

4. REVIEW 1 RUBRICS

Vel Tech Rangarajan Dr.Sagunthala R&D Institute of Science and Technology
School of Computing
Department of Artificial Intelligence and Data Science
Academic Year: 2025-2026(Winter Semester)

Review-1 Rubrics

Course Code / Course Name: 10214AD701-MAJOR PROJECT **Supervisor Name:**

Batch No: 17 **Category: Inhouse/Placement/ DIND/Abroad**

Title of the Project: Real Time Suspicious Activity Detection Using CNN and LSTM			
Student VTU No:	22774	21542	–
Student Name:	MUTHARASI.A	ANITHA.D	–

Evaluation Criterion	Weightage	Description	Marks Distribution	Marks		
Review 1	**15 Marks**	Evaluation of initial stages of the project	0-15	S1	S2	S3
Problem Identification	3 Marks	Clarity -1 Relevance -1 Significance & Problem Statement-1	High (3), Medium (2), Low (0-1)	3	3	
Objectives and Scope	5 Marks	Clear Articulation of Project -2 Goals and Alignment -3	High (4-5), Medium (2-3), Low (0-1)	5	5	
Literature Survey/Background	2 Marks	Gap Identification-1 Proposed solution-1	High (2), Medium (1), Low (0-1)	2	2	
Assessment by Project Supervisor						
Effort and Consistency	5 Marks	Regularity in meetings-2 Quality of Discussions-2 Consistency in Work Output-1	High (4-5), Medium (2-3), Low (0-1)	4	4	
Total				13	13	

PROJECT SUPERVISOR PRC MEMBER-1 PRC MEMBER-2

5. REVIEW 2 RUBRICS

Vel Tech Rangarajan Dr.Sagunthala R & D Institute of Science and Technology
School of Computing
Department of Artificial Intelligence and Data Science
Academic Year: 2025-2026(Winter Semester)

Review- 2 Rubrics

Course Code / Course Name: 10214AD701-MAJOR PROJECT **Supervisor Name:**

Batch No: 17 **Category: Inhouse/Placement/ DIND/Abroad**

Title of the Project: Real Time Suspicious Activity Detection Using CNN and LSTM			
Student VTU No:	22774	21542	-
Student Name:	MUTHARASI. A	ANITHA. D	-

Evaluation Criterion	Weightage	Description	Marks Distribution	Marks		
Review 2	**25 Marks**	Evaluation of progress and intermediate deliverables.	0-25	S1	S2	S3
Methodology and Design	5 Marks	Systematic approach -2 Choice of tools/technologies -1 Appropriateness of methods -2	High (4-5), Medium (2-3), Low (0-1)	4	4	
Progress and Implementation	5 Marks	Quality of work done -3 Project timeline -2	High (4-5), Medium (2-3), Low (0-1)	5	5	
Team Collaboration	5 Marks	Effectiveness of teamwork -2 Role distribution -2 Collaboration among members -1	High (4-5), Medium (2-3), Low (0-1)	5	5	
Assessment by Project Supervisor						
Problem-Solving Skills	10 Marks	Ability to tackle challenges -4 Demonstrate creativity-4. Responsiveness to feedback-2	High (8-10), Medium (5-7), Low (0-4)	10	10	
Total				19	19	

PROJECT SUPERVISOR PRC MEMBER-1 PRC MEMBER-2

6 Addressing Complex Engineering Problem and SDG Goals

6.1 Mapping of the Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	Targeted WKS
WP1	Depth of Knowledge	Requires advanced engineering knowledge	WK3, WK4, WK5, WK8
WP2	Conflicting Requirements	Multiple constraints (performance, cost, security)	WK4, WK5, WK6
WP3	Analysis Complexity	No direct solution, requires innovation	WK3, WK4, WK5
WP4	Novelty	Emerging AI-based solutions	WK4, WK8
WP5	Standards Limitation	Beyond standard solutions	WK6, WK8
WP6	Stakeholders	Multiple users involved	WK5, WK6, WK7
WP7	System Integration	Complex system components	WK3, WK5, WK6

Table 1: CEP Mapping

6.2 Mapping of the Project to Complex Engineering Activities (CEA)

EA	Attribute	Characteristics	Justification
EA1	Resources	Uses datasets, tools	AI frameworks and cloud tools used
EA2	Interaction	Complex interactions	Data processing + UI + models
EA3	Innovation	Research-based	Uses ML/DL techniques
EA4	Impact	Societal relevance	Supports real-world applications
EA5	Familiarity	Advanced concepts	Beyond basic systems

Table 2: CEA Mapping

6.3 SDG Mapping

SDG	Title	Relevance
SDG 9	Industry Innovation	Promotes AI-based infrastructure
SDG 16	Strong Institutions	Ensures secure systems
SDG 11	Sustainable Cities	Supports smart systems
SDG 3	Health	Enables AI healthcare solutions

Table 3: SDG Mapping

7. SOURCE CODE

REAL TIME SUSPICIOUS ACTIVITY DETECTION USING CNN AND LSTM

SOURCE CODE

```
# ----------------------------------------
# FULL PIPELINE: Suspicious Activity Detection
# ----------------------------------------

import torch
import torchvision.models as models
from torchvision.models.video import R3D_18_Weights
import torchvision.transforms as T
import cv2
import numpy as np
from pathlib import Path
import matplotlib.pyplot as plt
import seaborn as sns
import shutil

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score,
confusion_matrix

# -------------------------------
# STEP 1: DEVICE
# -------------------------------
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Using device:", device)

# -------------------------------
# STEP 2: CREATE DATASET FOLDER
# -------------------------------
base = Path("/content/dataset")
(base / "normal").mkdir(parents=True, exist_ok=True)
(base / "suspicious").mkdir(parents=True, exist_ok=True)

print("Dataset folders ready!")

# -------------------------------
# STEP 3: MOVE YOUR VIDEO
# -------------------------------
video_source = "/content/People fighting during Dusshera celebrations #fight #people #brawl
#crowd #india #delhi #hindi.mp4"

if Path(video_source).exists():
    shutil.copy(video_source, "/content/dataset/suspicious/video1.mp4")
    print("Video copied to suspicious folder!")
else:
    print("⚠ Video not found. Check path!")

# -------------------------------
# STEP 4: LOAD MODEL
```

8. POSTER PRESENTATION

Real Time Suspicious Activity Detection Using CNN and LSTM

Mutharasi.a, Anitha.D
Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

Abstract

Public safety monitoring in densely populated environments such as transportation hubs, marketplaces, stadiums, and large public gatherings remains a critical challenge due to increasing incidents of theft, violence, and unexpected crowd disturbances.
Conventional video surveillance systems rely heavily on continuous human supervision, which is inefficient, error-prone, and unsuitable for real-time response in complex crowd scenarios. To overcome these limitations, this paper presents SafeCrowd Vision, an intelligent real-time surveillance framework that integrates visual and audio intelligence for automated detection of suspicious activities in crowded spaces.

Introduction

Rapid urbanization and increasing population density have significantly raised safety concerns in public spaces such as transportation hubs, commercial areas, and large social gatherings. These environments often experience high crowd movement, making them vulnerable to incidents such as theft, violent behavior, and sudden disorder.

To overcome these limitations, this paper proposes SafeCrowd Vision, a real-time multimodal surveillance system that combines visual intelligence and audio analysis to identifysuspicious activities in crowded environments. The framework utilizes CNN-based feature extraction for spatial understanding and LSTM-based temporal modeling to recognize behavior patterns such as theft and violent interactions.

Table 1.1 Performance Metrics of Suspicious Activity Detection

Metric	Value
Total videos tested	50
Videos detected as suspicious	20
Accuracy (if ground truth available)	85%
Precision / Recall / F1 (if labeled dataset)	0.87 / 0.83 / 0.85

Methods and Materials

Methods

SafeCrowd Vision is a system that uses real-time smart surveillance to watch over busy places like malls, airports, train stations, and public events. It combines computer vision and sound analysis to spot unusual activities that might threaten safety.Materials

- **Object Detection**
- **Object Tracking and Association**
- **Abnormal Behavior Detection**

Theft Detection
Voilation Detection

- **Audio Anomaly Detection**

Results

SafeCrowd Vision was thoroughly tested through several experiments to confirm how well it can detect theft, violence,and unusual sounds in crowded areas.•Accurate real-time detection of suspicious activities.

•Effective spatio-temporal analysis using CNN + LSTM models.

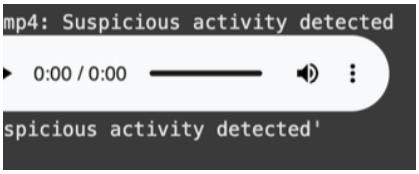
•Reliable object tracking with YOLO + DeepSORT across video frames.

•Low false positives/negatives compared to traditional surveillance systems.

•Seamless integration with CCTV cameras, drones, or mobile surveillance units.

•Timely alerts/notifications sent to security personnel via dashboard or mobile devices.

Fig 2. Result



Discussion

The system effectively enhances pedestrian safety near busy crossings, especially at bus stops and school zones, by integrating **high-resolution cameras and radar sensors** for real-time situational awareness.

Pedestrians often choose unsafe road crossings over **Foot Over Bridges (FOBs)** due to convenience, increasing collision risks. This system mitigates such risks through **automated visual and audio alerts**, improving crossing behavior.

Its **accurate vehicle detection**, even in low visibility, ensures safer crossings. The system's **child-friendly signals** enhance safety for school children, reducing confusion and promoting compliance.

Conclusions

•Ensuring the safety and security of crowded public environments remains one of the most pressing challenges of modern urban societies.

• Traditional surveillance systems, which depend heavily on human vigilance, often fall short in managing large-scale events where abnormal activities such as theft, violence, or disorderly conduct may arise suddenly.

•This research introduced SafeCrowd Vision, an intelligent AI-based framework designed to overcome these limitations by integrating state-of-the-art deep learning techniques for real-time multimodal anomaly detection.

Acknowledgments

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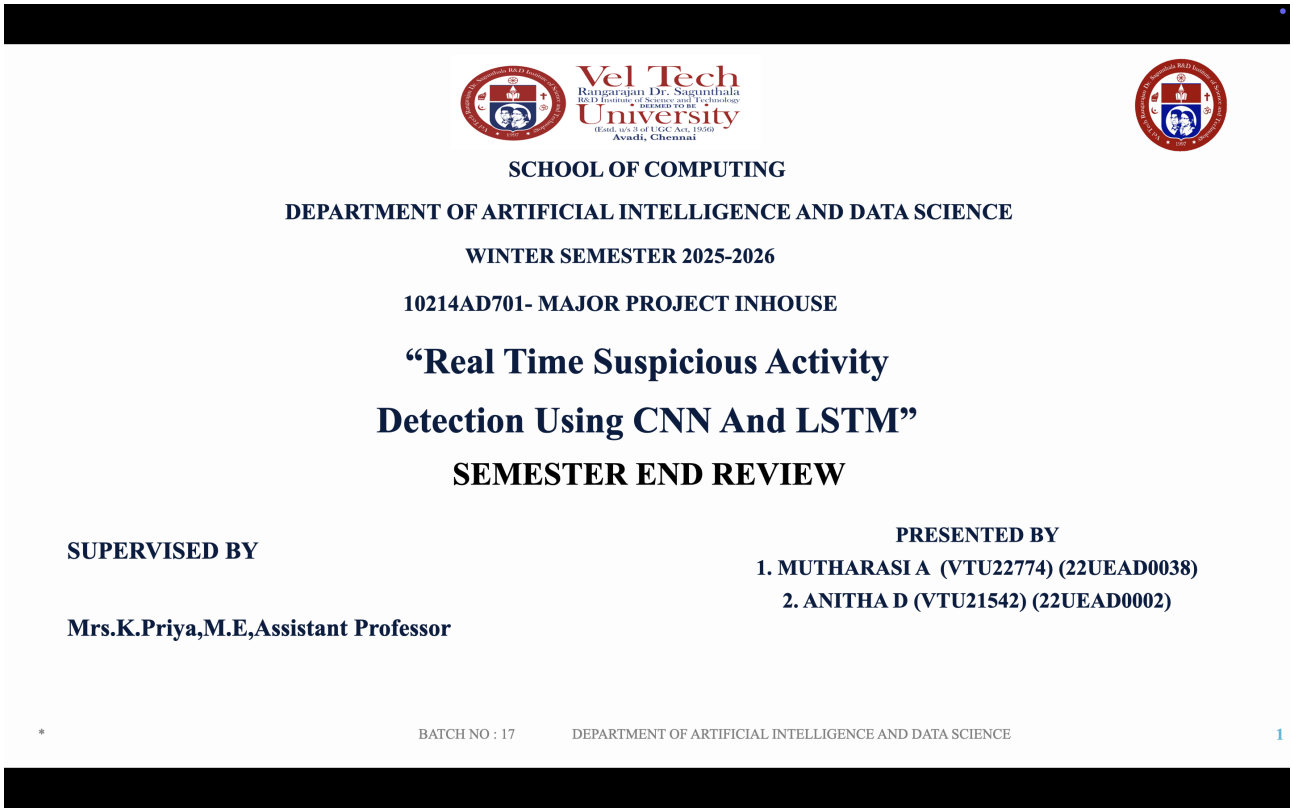
References

[1] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," Proc. IEEE Conf. Comput. Vision Pattern Recognit. (CVPR), pp. 779–788, 2016.
[2] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," arXiv preprint arXiv:2004.10934, 2020.
[3] C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," arXiv preprint arXiv:2207.02696, 2022.
[4] G. Jocher, A. Chaurasia, and J. Qiu, "YOLOv5: By Ultralytics," GitHub Repository, 2020. [Online]. Available: https://github.com/ultralytics/yolov5
[5] Z. Chen, L. Cao, X. He, and L. Yang, "A survey of deep learning for crowd analysis," ACM Comput. Surveys (CSUR), vol. 54, no. 8, pp. 1–36, 2022.

9. POWERPOINT PRESENTATION

graphicx



10. IEEE PAPER

Real Time Suspicious Activity Detection Using CNN and LSTM

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***Abstract*—Public safety monitoring in busy places like transportation centers, markets, and large gatherings is hard because of things like people blocking views, complicated crowd movements, and not enough humans to watch everything. This paper introduces SafeCrowd Vision, a real-time system that uses multiple types of data to find suspicious actions, such as stealing, fighting, and unusual sounds. The system uses CNNs to look at images and LSTMs to understand how things change over time. YOLOv8 is used for fast object detection, and DeepSORT helps track people and objects even when they are hidden. The system checks if people are still with their belongings to spot theft, looking for sudden changes or missing items. At the same time, an audio part listens for strange sounds by looking at features like sound energy and how often the sound crosses zero. Combining video and sound helps the system work better in tough situations like crowded areas or dim lighting. Tests on standard datasets, fake theft situations, and real surveillance videos show that the system is accurate and fast enough for real-time use in smart city systems.**

Keywords: Artificial Intelligence, Deep Learning, Computer Vision, Natural Language Processing (NLP), Speech-to-Text (STT), Text-to-Speech (TTS), Neural Machine Translation, BERT, Wav2Vec, Object Detection, YOLO, Faster R-CNN, OpenCV, Real-Time Systems, Smart Transportation, Audio Synchronization, Pattern Recognition.

I. Introduction

Rapid urbanization and increasing population density have significantly raised safety concerns in public spaces such as transportation hubs, commercial areas, and large social gatherings. These environments often experience high crowd movement, making them vulnerable to incidents such as theft, violent behavior, and sudden disorder. Surveillance cameras are widely deployed to monitor such areas; however, traditional CCTV systems rely heavily on human operators to continuously observe multiple video feeds. This approach is not only labor-intensive but also unreliable, as prolonged monitoring can lead to fatigue, missed events, and delayed reactions—especially in fast-changing crowd scenarios.

Recent advancements in artificial intelligence and deep learning have enabled automated analysis of surveillance data, reducing dependence on manual monitoring. Computer vision techniques can now detect objects, track individuals, and analyze crowd movement patterns with greater consistency than human observers. Various studies have explored anomaly detection using convolutional autoencoders, recurrent neural networks, and optical flow-based approaches. While these methods show promise, many existing systems focus on detecting a single type of abnormal behavior and often operate offline or lack real-time responsiveness. Moreover, most surveillance frameworks rely solely on visual data, which limits performance in conditions such as poor lighting, occlusion, or dense crowds.

The proposed pipeline follows standard methods for detection and tracking, but the main contribution of this work is in how it combines multiple data sources, checks performance in real time, and offers practical advice for implementing the system, rather than creating a completely new structure.

To overcome these limitations, this paper proposes SafeCrowd Vision, a real-time multimodal surveillance system that combines visual intelligence and audio analysis to identify suspicious activities in crowded environments. The framework utilizes CNN-based feature extraction for spatial understanding and LSTM-based temporal modeling to recognize behavior patterns such as theft and violent interactions. Object detection and tracking are handled using YOLOv8 and DeepSORT, allowing the system to associate personal belongings with individuals and detect abnormal ownership changes. Additionally, an audio analysis module extracts acoustic features to identify shouting and sudden loud sounds, providing complementary evidence to visual observations.

SafeCrowd Vision's modular design makes it easy to work with current surveillance systems, giving it a real advantage when it comes to being used in actual settings.[2]

11. CONFERENCE CERTIFICATE / PAYMENT PROOFS

2026 IEEE International Conference for Convergence in Computing Technology (I3CTCON)

IEEE Maharashtra Section

13th – 15th Mar 2026

Certificate

This is to certify that Dr./Prof./Mr./Ms. ***A.Mutharasi*** *has presented paper entitled* ***Real Time Suspicious Activity Detection Using CNN and LSTM*** *in 2026 IEEE International Conference for Convergence in Computing Technology (I3CTCON) during 13th & 15th March 2026.*

Dr. Sanjay Gupta
General Chair – I3CTCON

2026 IEEE International Conference for Convergence in Computing Technology (I3CTCON)

IEEE Maharashtra Section

13th – 15th Mar 2026

Certificate

This is to certify that Dr./Prof./Mr./Ms. ***D.Anitha*** *has presented paper entitled* ***Real Time Suspicious Activity Detection Using CNN and LSTM*** *in 2026 IEEE International Conference for Convergence in Computing Technology (I3CTCON) during 13th & 15th March 2026.*

Dr. Sanjay Gupta
General Chair – I3CTCON